(g) (1) $pq^2(-2p^2+4q) = -2p^3q^2+4pq^3$

(2) $(2x^3-5y^2)(3x^3-2y^2) = 6x^6-19x^3y^2+10y^4$

(3) $(2a+3b)^2 = (2a+3b)(2a+3b) = 4a^2+12ab+9b^2$

(4) $-3x(x-1)(x-4) = -3x(x^2-5x+4) = -3x^3+15x^2-12x$

(5) $2x(x-5)^2 = 2x(x^2-10x+25) = 2x^3-20x^2+50x$

(6) $(2x+5)(x^2-6x+4) = 2x^3-12x^2+8x+5x^2-30x+20$
$= 2x^3-7x^2-22x+20$

(7) $(x+3)(x^2-3x+9) = x^3+27$

(8) $(x+1)(x-2)(x-3) = (x+1)(x^2-5x+6)$
$= x^3-5x^2+6x+x^2-5x+6$
$= x^3-4x^2+x+6$

(9) $(x-2)^3 = (x-2)(x-2)(x-2) = (x-2)(x^2-4x+4)$
$= x^3-4x^2+4x-2x^2+8x-8$
$= x^3-6x^2+12x-8$

(10) $(x-1)(x+2)(x+1)(x+2) = (x-1)(x+1)(x+2)(x+2)$
$= (x^2-1)(x^2+4x+4)$
$= x^4+4x^3+4x^2-x^2-4x-4$
$= x^4+4x^3+3x^2-4x-4$

(11) $(x+1-y)(x+y-1) = (x-(y-1))(x+(y-1))$
$= x^2-(y-1)^2$
$= x^2-(y^2-2y+1)$
$= x^2-y^2+2y-1$

(12) $(x-y+1)^2 = (x-y+1)(x-y+1)$
$= [(x-y)+1][(x-y)+1]$
$= (x-y)^2+x-y+x-y+1$
$= x^2-2xy+y^2+2x-2y+1$

(13) $\left(\frac{3}{2}x-\frac{y}{3}\right)\left(\frac{4x}{3}-\frac{1}{2}y\right) = 2x^2-\frac{3}{4}xy-\frac{4}{9}xy+\frac{1}{6}y^2$
$= 2x^2-\frac{43}{36}xy+\frac{1}{6}y^2$

(14) $\left(\frac{2}{3}x-\frac{3}{2}y\right)^2 = \left(\frac{2}{3}x-\frac{3}{2}y\right)\left(\frac{2}{3}x-\frac{3}{2}y\right)$
$= \frac{4}{9}x^2-xy-xy+\frac{9}{4}y^2$
$= \frac{4}{9}x^2-2xy+\frac{9}{4}y^2$

(15) $\left(x+\frac{2}{x}\right)\left(x-\frac{2}{x}\right)=x^2-\frac{4}{x^2}$

(16) $\left(x+\frac{1}{x}\right)^3=\left(x+\frac{1}{x}\right)\left(x+\frac{1}{x}\right)\left(x+\frac{1}{x}\right)$

$=\left(x+\frac{1}{x}\right)\left(x^2+2+\frac{1}{x^2}\right)$

$=x^3+2x+\frac{1}{x}+x+\frac{2}{x}+\frac{1}{x^3}$

$=x^3+3x+\frac{3}{x}+\frac{1}{x^3}$

(17) $2(x-1)^2+x(x-1)(x+2)-(x-3)x^2$

$=2(x^2-2x+1)+x(x^2+x-2)-x^3+3x^2$

$=2x^2-4x+2+x^3+x^2-2x-x^3+3x^2$

$=6x^2-6x+2$

(18) $(x-3)(x+3)x+2x(x-2)^2-(x-1)(x^2+x+1)$

$=x(x^2-9)+2x(x^2-4x+4)-(x^3-1)$

$=x^3-9x+2x^3-8x^2+8x-x^3+1$

$=2x^3-8x^2-x+1$

(19) $\left[(x-y)(x^2+y^2)(x+y)\right]^2$

$=\left[(x-y)(x+y)(x^2+y^2)\right]^2$

$=[(x^2-y^2)(x^2+y^2)]^2$

$=(x^4-y^4)^2$

$=x^8-2x^4y^4+y^8$

(20) $(x-2)(x+2)(x^2-2x+4)(x^2+2x+4)$

$=(x-2)(x^2+2x+4)(x+2)(x^2-2x+4)$

$=(x^3-8)(x^3+8)$

$=x^6-64$

(21) $(2^n-3^n)(2^n+3^n)$

$=2^{2n}-3^{2n}$

(22) $(3^n-1)(3^{2n}+3^n+1)$

$=3^{3n}-1$

(23) $\left(x^m+y^{2m}\right)\left(x^m-3y^{2m}\right)$

$=x^{2m}-2x^my^{2m}-3y^{4m}$

(24) $\left(x^a-x^{-a}\right)^2$

$=(x^a-x^{-a})(x^a-x^{-a})$

$=x^{2a}-2\cdot x^a\cdot x^{-a}+x^{-2a}$

$=x^{2a}-2+x^{-2a}$

(h) (1) $8x^3y^2 - 6x^2y^3 + 2x^2y^2$
$= 2x^2y^2(4x - 3y + 1)$

(2) $3x^3 - 27x = 3x(x^2 - 9) = 3x(x - 3)(x + 3)$

(3) $x(x - 4) + 3 = x^2 - 4x + 3 = (x - 3)(x - 1)$

(4) $-x^3 + 2x^2 + 3x = -x(x^2 - 2x - 3)$
$= -x(x - 3)(x + 1)$

(5) $6a^2 + 17a + 12 = (3a + 4)(2a + 3)$

(6) $y^2 + \frac{5}{6}y - \frac{2}{3} = \frac{1}{6}(6y^2 + 5y - 4)$
$= \frac{1}{6}(3y + 4)(2y - 1)$

(7) $12x^2 - 4xy - 21y^2 = (6x + 7y)(2x - 3y)$

(8) $29xy - 15x^2 - 8y^2 = -15x^2 + 29xy - 8y^2$
$= -(15x^2 - 29xy + 8y^2)$
$= -(5x - 8y)(3x - y)$

(9) $x^3 + 27 = (x + 3)(x^2 - 3x + 9)$

(10) $24x^3y - 81y^4 = 3y(8x^3 - 27y^3)$
$= 3y(2x - 3y)(4x^2 + 6xy + 9y^2)$

(11) $13\frac{1}{2} + 4x^3 = \frac{27}{2} + 4x^3$
$= \frac{1}{2}(27 + 8x^3)$
$= \frac{1}{2}(3 + 2x)(9 - 6x + 4x^2)$

(12) $2^{3a} - 1 = (2^a - 1)(2^{2a} + 2^a + 1)$

(13) $x^4 - 16y^4 = (x^2 - 4y^2)(x^2 + 4y^2)$
$= (x - 2y)(x + 2y)(x^2 + 4y^2)$

(14) $4x^4 - 13x^2y^2 + 9y^4 = (4x^2 - 9y^2)(x^2 - y^2)$
$= (2x - 3y)(2x + 3y)(x - y)(x + y)$

(15) $(x^2 - 7x)^2 - 2x(x - 7) - 48$
$= (x^2 - 7x)^2 - 2(x^2 - 7x) - 48$
$= [(x^2 - 7x) + 6][(x^2 - 7x) - 8]$
$= (x^2 - 7x + 6)(x^2 - 7x - 8)$
$= (x - 6)(x - 1)(x - 8)(x + 1)$

(16) $(x^3 - 14y^3)^2 - 169y^6$
$= [(x^3 - 14y^3) - 13y^3][(x^3 - 14y^3) + 13y^3]$
$= (x^3 - 27y^3)(x^3 - y^3)$
$= (x - 3y)(x^2 + 3xy + 9y^2)(x - y)(x^2 + xy + y^2)$

(17) $x^2(p - 1) + 3x(-1 + p) + 4(1 - p)$
$= (p - 1)x^2 + (p - 1)3x - 4(p - 1)$
$= (p - 1)(x^2 + 3x - 4)$
$= (p - 1)(x + 4)(x - 1)$

(18) $a^2x^2 - a^2y^2 + 9y^2 - 9x^2$
$= a^2(x^2 - y^2) - 9(x^2 - y^2)$
$= (x^2 - y^2)(a^2 - 9)$
$= (x - y)(x + y)(a - 3)(a + 3)$

(19) $a^3 - a - 1 + a^2$
$= a(a^2 - 1) + (a^2 - 1)$
$= (a^2 - 1)(a + 1)$
$= (a - 1)(a + 1)(a + 1)$

(20) $a^3 + a^2 + b^3 - b^2$
$= a^3 + b^3 + a^2 - b^2$
$= (a + b)(a^2 - ab + b^2) + (a - b)(a + b)$
$= (a + b)(a^2 - ab + b^2 + a - b)$

(21) $(2x - 1)^2 - a + 4ax^2$
$= (2x - 1)^2 + a(4x^2 - 1)$
$= (2x - 1)^2 + a(2x - 1)(2x + 1)$
$= (2x - 1)[(2x - 1) + a(2x + 1)]$
$= (2x - 1)(2x - 1 + 2ax + a)$

(22) $y^2 - 6y - x^2 + 9$
$= y^2 - 6y + 9 - x^2$
$= (y - 3)(y - 3) - x^2$
$= (y - 3)^2 - x^2$
$= (y - 3 - x)(y - 3 + x)$

(23) $1 - p^2 - 4pq - 4q^2$
$= 1 - (p^2 + 4pq + 4q^2)$
$= 1 - (p + 2q)(p + 2q)$
$= 1 - (p + 2q)^2$
$= [1 - (p + 2q)][1 + (p + 2q)]$
$= (1 - p - 2q)(1 + p + 2q)$

(24) $ax^2 - 2ax + a + bx^2 - b$
$= a(x^2 - 2x + 1) + b(x^2 - 1)$
$= a(x - 1)(x - 1) + b(x - 1)(x + 1)$
$= (x - 1)[a(x - 1) + b(x + 1)]$
$= (x - 1)(ax - a + bx + b)$

(25) $a - c + 2b + cx - ax - 2bx$

$= a - ax + 2b - 2bx + cx - c$

$= a(1 - x) + 2b(1 - x) - c(1 - x)$

$= (1 - x)(a + 2b - c)$

(26) $a^2x + 3x^2 + 3a^2 + 9x - x^3 - 27$

$= a^2x + 3a^2 + 3x^2 + 9x - x^3 - 27$

$= a^2(x + 3) + 3x(x + 3) - (x + 3)(x^2 - 3x + 9)$

$= (x + 3)(a^2 + 3x - x^2 + 3x - 9)$

$= (x + 3)[a^2 - (x^2 - 6x + 9)]$

$= (x + 3)[a^2 - (x - 3)(x - 3)]$

$= (x + 3)[a^2 - (x - 3)^2]$

$= (x + 3)[a - (x - 3)][a + (x - 3)]$

$= (x + 3)(a - x + 3)(a + x - 3)$

(i) (1) $\dfrac{x^2 - 4}{x^2 + 2x} \times \dfrac{x^2 - 9}{x^2 - 5x + 6}$

$= \dfrac{(x - 2)(x + 2)}{x(x + 2)} \times \dfrac{(x - 3)(x + 3)}{(x - 3)(x - 2)}$

$= \dfrac{x + 3}{x}$

(2) $\dfrac{x^4 + 27x}{2 - 3x} \div \dfrac{x^3 - 3x^2 + 9x}{3x^2 + x - 2}$

$= \dfrac{x(x^3 + 27)}{-(3x - 2)} \times \dfrac{(3x - 2)(x + 1)}{x(x^2 - 3x + 9)}$

$= \dfrac{x(x + 3)(x^2 - 3x + 9)}{-(3x - 2)} \times \dfrac{(3x - 2)(x + 1)}{x(x^2 - 3x + 9)}$

$= -(x + 3)(x + 1)$

(3) $\dfrac{x^2 + x - 6}{3x^2 - 12x} \div \dfrac{x^3 - 2x^2}{x^2 - 16} \times \dfrac{1}{x + 4}$

$= \dfrac{(x + 3)(x - 2)}{3x(x - 4)} \times \dfrac{(x - 4)(x + 4)}{x^2(x - 2)} \times \dfrac{1}{x + 4}$

$= \dfrac{x + 3}{3x^3}$

(4) $\dfrac{2x^2y + 2x}{x^2y^2 - 1} \div \dfrac{y}{3x^2 - 3x} \div \dfrac{6x^2}{xy^2 - y}$

$= \dfrac{2x(xy + 1)}{(xy + 1)(xy - 1)} \times \dfrac{3x(x - 1)}{y} \times \dfrac{y(xy - 1)}{6x^2}$

$= x - 1$

(5) $\dfrac{x+3}{3x}+\dfrac{3y+1}{6y^2}-\dfrac{2y+x}{2xy}$ LCD $= 6xy^2$

$$=\frac{2y^2(x+3)}{6xy^2}+\frac{x(3y+1)}{6xy^2}-\frac{3y(2y+x)}{6xy^2}$$

$$=\frac{2y^2(x+3)+x(3y+1)-3y(2y+x)}{6xy^2}$$

$$=\frac{2xy^2+6y^2+3xy+x-6y^2-3xy}{6xy^2}$$

$$=\frac{2xy^2+x}{6xy^2}$$

$$=\frac{x(2y^2+1)}{6xy^2}$$

$$=\frac{2y^2+1}{6y^2}$$

(6) $\dfrac{1}{x-2}+\dfrac{1}{x+2}-\dfrac{2x}{x^2+4}$ LCD $= (x-2)(x+2)(x^2+4)$

$$=\frac{(x+2)(x^2+4)+(x-2)(x^2+4)-2x(x-2)(x+2)}{(x-2)(x+2)(x^2+4)}$$

$$=\frac{x^3+4x+2x^2+8+x^3+4x-2x^2-8-2x(x^2-4)}{(x-2)(x+2)(x^2+4)}$$

$$=\frac{2x^3+8x-2x^3+8x}{(x-2)(x+2)(x^2+4)}$$

$$=\frac{16x}{(x-2)(x+2)(x^2+4)}$$

(7) $\dfrac{1}{x-2}-\dfrac{4}{(x+2)^2}-\dfrac{1}{x+2}$ LCD $= (x-2)(x+2)^2$

$$=\frac{(x+2)^2-4(x-2)-(x-2)(x+2)}{(x-2)(x+2)^2}$$

$$=\frac{x^2+4x+4-4x+8-(x^2-4)}{(x-2)(x+2)^2}$$

$$=\frac{x^2+12-x^2+4}{(x-2)(x+2)^2}$$

$$=\frac{16}{(x-2)(x+2)^2}$$

(8) $\dfrac{x-3}{x-2}+\dfrac{x-2}{3-x}+\dfrac{3x-8}{x^2-5x+6}$

$=\dfrac{x-3}{x-2}-\dfrac{x-2}{x-3}+\dfrac{3x-8}{(x-3)(x-2)}$

$=\dfrac{(x-3)(x-3)-(x-2)(x-2)+3x-8}{(x-2)(x-3)}$ LCD $=(x-2)(x-3)$

$=\dfrac{x^2-6x+9-(x^2-4x+4)+3x-8}{(x-2)(x-3)}$

$=\dfrac{x^2-3x+1-x^2+4x-4}{(x-2)(x-3)}$

$=\dfrac{x-3}{(x-2)(x-3)}$

$=\dfrac{1}{(x-2)}$

(9) $\dfrac{3}{1-x}+\dfrac{3x^2+6x}{3x^2+6}\times\dfrac{2+x^2}{x^3-x^2}$

$=\dfrac{-3}{x-1}+\dfrac{3x(x+2)}{3(x^2+2)}\times\dfrac{(x^2+2)}{x^2(x-1)}$

$=\dfrac{-3}{x-1}+\dfrac{x+2}{x(x-1)}$ LCD $=x(x-1)$

$=\dfrac{-3x+(x+2)}{x(x-1)}$

$=\dfrac{-3x+x+2}{x(x-1)}$

$=\dfrac{-2x+2}{x(x-1)}$

$=\dfrac{-2(x-1)}{x(x-1)}$

$=\dfrac{-2}{x}$

(10) $\dfrac{x}{1-4x^2}\div\left(\dfrac{3x}{4x^2-9}-\dfrac{2x}{4x^2-4x-3}\right)$

$=\dfrac{x}{1-4x^2}\div\left(\dfrac{3x}{(2x-3)(2x+3)}-\dfrac{2x}{(2x-3)(2x+1)}\right)$

$=\dfrac{x}{1-4x^2}\div\dfrac{3x(2x+1)-2x(2x+3)}{(2x-3)(2x+3)(2x+1)}$

$=\dfrac{x}{(1-2x)(1+2x)}\times\dfrac{(2x-3)(2x+3)(2x+1)}{6x^2+3x-4x^2-6x}$

$=\dfrac{x}{(1-2x)}\times\dfrac{(2x-3)(2x+3)}{2x^2-3x}$

$=\dfrac{x}{(1-2x)}\times\dfrac{(2x-3)(2x+3)}{x(2x-3)}$

$=\dfrac{2x+3}{1-2x}$

(11) $\dfrac{\frac{x}{y}-\frac{25y}{x}}{\frac{5y}{x^2}+\frac{1}{x}}$

$$=\left(\frac{x}{y}-\frac{25y}{x}\right)\div\left(\frac{5y}{x^2}+\frac{1}{x}\right)$$

$$=\left(\frac{x^2-25y^2}{xy}\right)\div\left(\frac{5y+x}{x^2}\right)$$

$$=\frac{(x-5y)(x+5y)}{xy}\times\frac{x^2}{x+5y}$$

$$=\frac{x(x-5y)}{y}$$

(12) $\dfrac{\frac{7}{9x^2-4}}{\frac{2}{2x+1}-\frac{3}{3x-2}}$

$$=\left[\frac{7}{(3x-2)(3x+2)}\right]\div\left[\frac{2}{2x+1}-\frac{3}{3x-2}\right]$$

$$=\frac{7}{(3x-2)(3x+2)}\div\left[\frac{2(3x-2)-3(2x+1)}{(2x+1)(3x-2)}\right]$$

$$=\frac{7}{(3x-2)(3x+2)}\times\frac{(2x+1)(3x-2)}{6x-4-6x-3}$$

$$=\frac{7}{(3x+2)}\times\frac{(2x+1)}{-7}$$

$$=\frac{-(2x+1)}{(3x+2)}$$

(j)* $x+\dfrac{1}{x}=5$

(1) $\left(x+\dfrac{1}{x}\right)^2=(5)^2$

$\therefore x^2+2+\dfrac{1}{x^2}=25$

$\therefore x^2+\dfrac{1}{x^2}=23$

(2) $x^3+\dfrac{1}{x^3}=\left(x+\dfrac{1}{x}\right)\left(x^2-1+\dfrac{1}{x^2}\right)$

$=(5)(23-1)$

$=5\times 22$

$=110$

(3) $\left(x-\dfrac{1}{x}\right)^2=x^2-2+\dfrac{1}{x^2}$

$=x^2+\dfrac{1}{x^2}-2$

$=23-2$

$=21$

$\therefore x-\dfrac{1}{x}=\sqrt{21}$

(4) $x^2 - \frac{1}{x^2} = \left(x - \frac{1}{x}\right)\left(x + \frac{1}{x}\right)$
$= (\sqrt{21})(5)$
$= 5\sqrt{21}$

(k)* $x + y = 2$ and $x^2 + y^2 = 8$

(1) $(x + y)^2 = (2)^2$
$\therefore x^2 + 2xy + y^2 = 4$
$\therefore x^2 + y^2 + 2xy = 4$
$\therefore 8 + 2xy = 4$
$\therefore 2xy = -4$
$\therefore xy = -2$

(2) $x^3 + y^3 = (x + y)(x^2 - xy + y^2)$
$= (2)(8 - (-2))$
$= (2)(10)$
$= 20$

(3) $(x - y)^2 = x^2 - 2xy + y^2$
$= x^2 + y^2 - 2(xy)$
$= 8 - 2(-2)$
$= 12$
$\therefore x - y = \sqrt{12}$

(4) $x^4 - y^4 = (x^2 - y^2)(x^2 + y^2)$
$= (x - y)(x + y)(x^2 + y^2)$
$= (\sqrt{12})(2)(8)$
$= 16(\sqrt{12})$

(l)* $a = x^2 + y^2$
$b = (x + y)^2 = x^2 + 2xy + y^2 = x^2 + y^2 + 2xy$
$c = (x - y)^2 = x^2 - 2xy + y^2 = x^2 + y^2 - 2xy$

(1) $b\ ;\ a\ ;\ c$

(2) $c\ ;\ a\ ;\ b$

(m)* $(ac + bd)^2 + (ad - bc)^2$
$= a^2c^2 + 2abcd + b^2d^2 + a^2d^2 - 2abcd + b^2c^2$
$= a^2c^2 + a^2d^2 + b^2c^2 + b^2d^2$
$= a^2(c^2 + d^2) + b^2(c^2 + d^2)$
$= (c^2 + d^2)(a^2 + b^2)$
$= (6)(7)$
$= 42$

(n)* $999\,999\,999\,999^2 - 1$
$= (999\,999\,999\,999 - 1)(999\,999\,999\,999 + 1)$
$= (999\,999\,999\,998)(1\,000\,000\,000\,000)$
$= 999\,999\,999\,998\,000\,000\,000\,000$
Sum of digits: $(9 \times 11) + 8 = 107$

(o)** (1) Let the numbers be x and $x + 2$

Even number between them is $x + 1$

Product of odd numbers: $x(x + 2) = x^2 + 2x$

Square of even number: $(x + 1)^2 = x^2 + 2x + 1$

Conclusion: Product of odd numbers is one less than the square of even number.

(2) Let the tens digit be x and the units digits be y.

$\therefore$ The number is: $10x + y$

$$\therefore (10x + y)(11) = (10x + y)(10 + 1)$$
$$= 100x + 10x + 10y + y$$
$$= 100x + 10(x + y) + y$$

Conclusion: The hundreds of the result is x and the tens is $(x + y)$ and the units is y.

(3) Let the tens digit be x and the units be 5

$\therefore$ The number is $10x + 5$

$$\therefore (10x + 5)^2 = 100x^2 + 100x + 25$$
$$= 100x(x + 1) + 20 + 5$$
$$= 100x(x + 1) + (2 \times 10) + 5$$

Conclusion: The hundreds digit is $x(x + 1)$, the tens digit is 2 and the units 5.